

Data Science with Generative AI

Program Outline

The Data Science course equips learners with skills to extract insights, make data-driven decisions, and solve real-world problems. Starting with Python fundamentals and statistics, it covers data manipulation (NumPy, Pandas), visualization (Matplotlib, Seaborn), and database management (SQL, NoSQL).

Key topics include machine learning (regression, decision trees, SVMs, Random Forests), deep learning (neural networks, CNNs, RNNs, GANs), and NLP (BERT, GPT). Learners gain hands-on experience with tools like TensorFlow, PyTorch, Hugging Face, and LangChain, tackling projects such as customer segmentation, image classification, and text summarization.

The course also explores cutting-edge AI topics like LLMs, prompt engineering, and RAG, ensuring mastery of the end-to-end data science workflow for a successful career in AI.

Learning Objective

- Develop a strong foundation in Python programming, including data types, loops, conditionals, functions, and object-oriented programming.
- Understand and work with data structures like lists, tuples, dictionaries, and sets for data organization and manipulation.
- Gain expertise in advanced Python concepts, including iterators, generators, lambda functions, and functional programming tools like `map`, `reduce`, and `filter`.
- Learn object-oriented programming principles such as inheritance, polymorphism, encapsulation, and abstraction, and implement advanced features like decorators and dunder methods.
- Master file handling, exception handling, debugging, and logging in Python while exploring multithreading and multiprocessing.
- Use industry-standard libraries such as NumPy, Pandas, Matplotlib, Seaborn, and Plotly for data analysis and visualization.
- Build RESTful APIs with Flask, including creating, deploying, and securing APIs, along with user authentication and testing.
- Work with SQL and NoSQL databases, performing CRUD operations, schema design, optimization, and integration with tools like MongoDB.
- Apply statistical concepts like probability distributions, hypothesis testing, correlation, and variance using Python to analyze real-world datasets.
- Learn machine learning fundamentals, including supervised and unsupervised learning, and implement regression, classification, and clustering models.
- Perform exploratory data analysis (EDA) on datasets, identifying trends and insights through practical case studies and projects.

- Understand and implement linear and logistic regression, regularization techniques, and evaluation metrics such as R^2 , MSE, and MAE.
- Explore advanced machine learning techniques, including decision trees, SVMs, Naive Bayes, ensemble methods, and clustering algorithms.
- Dive into anomaly detection and time series analysis, using techniques like ARIMA, SARIMA, and machine learning models for forecasting and anomaly detection.
- Build and deploy deep learning models for tasks like image classification and sentiment analysis using frameworks like TensorFlow and PyTorch.
- Gain hands-on experience with NLP techniques, including text preprocessing, embeddings, and transformer models like BERT and GPT.
- Explore generative AI concepts such as autoencoders, GANs, and large language models (LLMs) for advanced applications in AI and natural language understanding.
- Learn object detection, image segmentation, and advanced frameworks like YOLO, Detectron2, and TFOOD for building AI-powered solutions.
- Implement and deploy cutting-edge projects like AI assistants, sentiment analysis tools, and recommendation systems.
- Master fine-tuning, prompt engineering, and Retrieval-Augmented Generation (RAG) for customizing AI models to specific use cases.

Learning Outcomes:

- Develop a strong foundation in Python programming, including object-oriented principles, data structures, and file handling for building scalable and efficient applications.
- Master statistical analysis, probability distributions, and hypothesis testing to analyze data and draw meaningful insights for decision-making.
- Gain expertise in machine learning techniques, including regression, classification, ensemble methods, and clustering, while working on real-world datasets and projects.
- Build and deploy deep learning models using frameworks like TensorFlow and PyTorch, implementing advanced architectures such as CNNs, RNNs, and GANs.
- Design and implement RESTful APIs using Flask to enable seamless communication between applications and systems.
- Leverage databases, both SQL and NoSQL (e.g., MongoDB), for efficient data storage, querying, and optimization, with best practices for schema design and indexing.
- Explore and apply data visualization tools such as Matplotlib, Seaborn, and Plotly to create meaningful and interactive dashboards.
- Understand and utilize modern NLP techniques, including transformer-based models like BERT and GPT, for text processing, sentiment analysis, and summarization.
- Implement generative AI techniques for creative applications such as text generation, music composition, and machine translation, while addressing ethical concerns.
- Employ Gen AI tools like Langchain, LlamaIndex, and VectorDB for building cutting-edge AI-driven applications.

- Design and optimize end-to-end pipelines for machine learning and deep learning projects, integrating concepts such as feature engineering, model training, and evaluation.
- Develop and fine-tune large language models (LLMs) for domain-specific applications and enhance them using prompt engineering and retrieval-augmented generation (RAG) techniques.
- Build and deploy sophisticated object detection, tracking, and segmentation solutions using frameworks like YOLO, Detectron2, and TFOD.
- Utilize cloud infrastructure and DevOps tools for deployment and monitoring, ensuring secure, reliable, and scalable applications.
- Complete industry-relevant projects, such as anomaly detection, recommendation systems, and API development, showcasing the ability to apply theoretical knowledge to practical use cases.

Tools & Technologies

- **Python:** Core programming language for development and automation.
- **Python IDEs:** PyCharm, VS Code, Jupyter Notebook, Google Colab, Deepnote
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, Plotly, Bokeh.
- **Databases:** MySQL, MongoDB
- **API Development & Testing:** Flask, Postman
- **Machine Learning & Deep Learning:**
 - **Frameworks:** Scikit-learn, TensorFlow, Keras, PyTorch.
 - **Libraries:** XGBoost, CatBoost, Hugging Face, OpenCV.
- **Computer Vision:** YOLOv9, Mask R-CNN, Detectron2, OpenCV, Roboflow.
- **Natural Language Processing (NLP):**
 - **Libraries:** NLTK, spaCy, TextBlob.
 - **Tools:** LangChain, LlamaIndex, Hugging Face Hub.
- **Generative AI:**
 - **Frameworks:** GPTs, Autoencoders, LangChain, ChromaDB.
 - **Deployment Tools:** Streamlit

Curriculum

Milestone 1: Python | Week 1-7

Milestone Overview

This milestone introduces Python programming fundamentals, covering core concepts, data handling, object-oriented programming (OOP), file management, APIs, and databases. It provides a solid foundation in Python development, culminating in hands-on projects that simulate real-world applications.

Comprehensive List of Topics:

- **Python Basics:** Covers Python's features, variables, data types, operators, expressions, and control flow statements like loops and conditionals.
- **Data Types & Structures:** Explores strings, lists, tuples, dictionaries, and sets, focusing on their properties, manipulation techniques, and practical applications.
- **Functions:** Introduces function definition, calling, arguments, return values, lambda functions, scope, and recursion for writing reusable and efficient code.
- **Object-Oriented Programming (OOPs):** Covers classes, objects, inheritance, polymorphism, encapsulation, and abstraction to implement modular and reusable code structures.
- **Files, Exception Handling, Logging, & Memory Management:** Focuses on reading/writing files, handling exceptions, implementing logging for debugging, and understanding garbage collection and memory optimization.
- **Data Toolkit:** Introduces NumPy for numerical computations, Pandas for data handling, and Matplotlib/Seaborn for data visualization techniques.
- **RESTful API & Flask:** Covers API concepts, building lightweight web applications with Flask, creating RESTful API endpoints, and implementing authentication mechanisms.
- **Databases & MongoDB:** Explores SQL vs NoSQL databases, working with SQLite/MySQL, understanding MongoDB, and performing CRUD operations efficiently.

Expected Learning Outcomes

- Gain proficiency in Python programming fundamentals.
- Understand data structures and their applications.
- Develop modular and reusable code using functions.
- Implement object-oriented principles for efficient coding.
- Handle files, exceptions, and logs proficiently.
- Use NumPy and Pandas for data analysis.
- Build and deploy RESTful APIs using Flask.
- Work with SQL and NoSQL databases, including MongoDB.

Project

Project	Description
Advance Review Scraper	This project automates extracting product reviews, ratings, and comments from Flipkart to analyze customer sentiment and product performance.

Milestone 2: Statistics | Week 8-10

Milestone description

This milestone covers fundamental and advanced statistical concepts essential for data analysis and decision-making. It includes probability distributions, hypothesis testing, ANOVA, correlation, and statistical implementations in Python. By the end of this milestone, learners will gain the ability to apply statistical methods to real-world datasets, perform hypothesis testing, and make data-driven decisions.

Comprehensive List of Topics:

- **Statistics Basics** – Covers the role of statistics in data science, differentiates between descriptive and inferential statistics, explores types of data and sampling techniques, and explains levels of measurement.
- **Measures of Central Tendency & Dispersion** – Introduces mean, median, mode, range, variance, and standard deviation, along with Python implementations to analyze data spread and variability.
- **Measure of Symmetry (Skewness) & Variability** – Explains skewness (left, right, and symmetric distributions) and standard deviation, variance, and implementation of spread calculations in Python.
- **Set Theory & Correlation** – Covers fundamental set operations, covariance, correlation, and Python implementations to measure relationships between variables.
- **Probability & Random Variables** – Explores random variables, probability distributions, and key concepts like probability mass function (PMF), probability density function (PDF), and cumulative density function (CDF).
- **Discrete & Continuous Probability Distributions** – Introduces binomial, Bernoulli, Poisson, discrete and continuous uniform distributions, and normal (Gaussian) distribution with real-world applications.
- **Central Limit Theorem & Estimation** – Explains how sample means approximate normal distribution, point and interval estimation, confidence intervals, and margin of error.
- **Hypothesis Testing & Statistical Inference** – Covers Z-tests, T-tests, hypothesis testing mechanisms, p-values, Type 1 & Type 2 errors, and when to use T-test vs. Z-test.
- **Chi-Square, F-Distribution, & ANOVA** – Discusses chi-square tests, Bayes' theorem, goodness of fit, F-distribution, F-test, ANOVA types, partitioning variance, and their assumptions.
- **Python Implementation for Statistics** – Applies statistical methods in Python, including hypothesis testing, correlation, regression, and ANOVA, to automate data analysis and decision-making.

Expected Learning Outcomes

- Understand fundamental and advanced statistical concepts.
- Perform data sampling, distribution analysis, and statistical inference.

- Implement statistical methods such as hypothesis testing, ANOVA, and correlation.
- Apply probability distributions to model real-world data scenarios.
- Gain proficiency in Python for statistical analysis and visualization.
- Develop a strong foundation in data-driven decision-making using statistics.

Milestone 3: Machine Learning | Week 11-20

Milestone description

This milestone provides a comprehensive journey through Machine Learning (ML), covering fundamental concepts, exploratory data analysis (EDA), regression and classification models, ensemble techniques, dimensionality reduction, clustering, anomaly detection, and time series forecasting. Learners will gain hands-on experience in building, evaluating, and optimizing ML models using Python.

Comprehensive List of Topics

- **Introduction to Machine Learning** – Covers the fundamentals of ML, including AI vs ML vs DL vs DS, types of ML, data partitioning (train, test, validation), and concepts like overfitting, underfitting, and bias-variance tradeoff.
- **Feature Engineering & Data Preprocessing** – Focuses on handling missing data, imbalanced datasets, outliers, feature extraction, scaling, and encoding techniques to improve model performance, with hands-on implementation.
- **Exploratory Data Analysis (EDA)** – Covers data visualization, statistical analysis, and extracting meaningful insights from datasets, including projects like Flight Price Prediction and Travel Data Analysis.
- **Linear Regression & Evaluation Metrics** – Introduces simple and multiple linear regression, polynomial regression, evaluation metrics (R^2 , Adjusted R^2 , MSE, MAE, RMSE), ML pipelines, and model deployment.
- **Logistic Regression & Classification Metrics** – Covers logistic regression, regularization, cross-validation, hyperparameter tuning, classification metrics (Precision, Recall, F1-score, ROC-AUC), and an Email Spam Classification project.
- **Decision Trees & Random Forest** – Explains decision tree classification, pruning techniques, random forest classifier and regressor, and includes a Car Evaluation project.
- **Support Vector Machines (SVM) & Naive Bayes** – Covers SVM soft/hard margins, kernel tricks, Naive Bayes classifiers, and their variants, with an emergency survival prediction project.
- **Ensemble Learning: Bagging & Boosting** – Introduces ensemble learning with bagging (Random Forest) and boosting techniques (AdaBoost, Gradient Boosting, XGBoost), with a Sentiment Analysis project.

- **KNN & Stacking Models** – Covers K-Nearest Neighbors for classification and regression, stacking techniques, meta-learners, and an Energy Consumption Forecasting project using stacked models.
- **Dimensionality Reduction & Clustering** – Introduces feature selection, PCA, and clustering methods (K-Means, Hierarchical, DBSCAN) with a Movie Recommendation Clustering project.
- **Anomaly Detection & Time Series Forecasting** – Covers isolation forest, local outlier factors, DBSCAN for anomalies, time series forecasting (AR, MA, ARIMA, SARIMA, SARIMAX), and a Stock Market Anomaly Detection project.

Expected Learning Outcomes

- Understand the fundamentals of machine learning and its key applications.
- Perform data preprocessing, feature engineering, and EDA for better model training.
- Implement regression, classification, clustering, and ensemble models in Python.
- Optimize models using hyperparameter tuning and cross-validation.
- Work with dimensionality reduction, anomaly detection, and time series forecasting.
- Gain hands-on experience through real-world projects in multiple domains.

Projects	Description
Sensor Fault Detection	A wafer is a thin semiconductor slice (e.g., c-Si) used in ICs and solar cells. It undergoes microfabrication (doping, etching, deposition, photolithography) before dicing and packaging into integrated circuits.
Shipment Price Prediction	The Shipment Price Prediction project uses advanced machine learning to accurately forecast shipping costs, helping businesses optimize logistics, reduce expenses, and enhance efficiency.
Spam detection	This project builds a machine learning system to classify text as spam or ham, enhancing email and SMS filtering. It helps detect unwanted messages for improved communication security.
Visibility distance prediction	This project develops a machine learning model to predict maximum visibility based on weather and geographical factors. It enhances safety and efficiency in aviation, transportation, and outdoor activities.

Milestone 4: Deep Learning | Week: 21-29

Milestone description

This milestone provides a comprehensive deep learning curriculum, covering foundational concepts, neural network architectures, image classification, object detection, segmentation, tracking, and generative models. Learners will gain hands-on experience with TensorFlow, PyTorch, CNNs, YOLO, Detectron2, TFOD2, and GANs, developing state-of-the-art AI models for various real-world applications.

Comprehensive List of Topics

- **Neural Network & Perceptron** – Covers deep learning fundamentals, neural network architectures, activation functions, loss functions, optimization techniques, forward & backward propagation, gradient issues, and visualization, with hands-on implementation.
- **Deep Learning Frameworks (TensorFlow & PyTorch)** – Focuses on installing and working with TensorFlow and PyTorch, building neural networks, debugging using TensorBoard and Netron, cloud training in Colab Pro, and implementing a sentiment analysis project.
- **CNN & Image Classification** – Explores convolutional layers, pooling, feature extraction, training CNNs from scratch, and deploying image classification models as web apps.
- **Advanced CNN Architectures** – Covers deep CNN variants (LeNet-5, AlexNet, GoogLeNet, VGGNet, ResNet, Inception), transfer learning, and visualization.
- **Object Detection: RCNN & YOLO** – Introduces region-based CNN (Fast & Faster RCNN) and real-time object detection using YOLOv9, covering data annotation, model training, and inference.
- **Object Detection: Detectron2 & TFOD2** – Focuses on implementing Detectron2 and TensorFlow Object Detection API (TFOD2) for advanced object detection tasks.
- **Image Segmentation & Instance Segmentation** – Covers scene understanding, types of segmentation, Mask R-CNN, and transitioning from bounding boxes to polygon masks.

- **Object Tracking** – Introduces dataset annotation, Kalman filters, YOLO-based tracking, and DeepSORT for real-time object tracking applications.
- **Generative Adversarial Networks (GANs)** – Covers GAN architecture, discriminator and generator networks, WGANs, DCGANs, StyleGANs, and synthetic data generation.

Expected Learning Outcomes

- Develop a strong foundation in neural networks and deep learning frameworks like TensorFlow & PyTorch.
- Implement CNNs, RCNN, YOLO, and Detectron2 for image classification, object detection, and segmentation.
- Work with Mask R-CNN for instance segmentation and DeepSORT for object tracking.
- Train GANs for synthetic data generation and creative AI applications.
- Gain hands-on experience through real-world projects in computer vision and deep learning.

Project	Description
Audio Classification System	This project builds an audio binary classification model to classify speech. Users can input an audio file to receive a spectrogram image and target label. The dataset includes 6,000 cat and dog audio files.
Helmet detection	This project uses computer vision to detect helmets in real-time, enhancing safety compliance in workplaces and traffic monitoring. It ensures accurate identification to prevent violations and accidents.
AI Driven Text Summarizer	This project develops an AI-driven text summarizer that automatically condenses long texts while preserving key information. It enhances readability and efficiency for users by generating concise summaries.
Object Detection	This project develops an object detection system using AI to identify and locate objects in images or videos. It enhances automation, security, and real-time monitoring applications.

Milestone 5: NLP | Week: 30-32

Milestone description

This milestone introduces learners to Natural Language Processing (NLP) and covers essential

techniques for text processing, useful NLP libraries, and advanced models like attention mechanisms and transformers. Learners will gain hands-on experience in building models for text categorization, emotion detection, and text summarization using popular frameworks and architectures.

Comprehensive List of Topics

- **NLP Introduction & Text Processing** – Covers the fundamentals of NLP, computational linguistics, text processing techniques, regex, tokenization, normalization, word embeddings (Word2Vec, Doc2Vec), and vector-based representations, with hands-on implementation in news categorization.
- **Useful NLP Libraries & Networks** – Focuses on NLP libraries (NLTK, SpaCy, TextBlob, Stanford NLP), neural network architectures (RNNs, LSTMs, Bi-LSTMs, GRUs), and real-world NLP model implementations, with an emotion detection project using Bi-LSTM.
- **Attention-Based Models & Transfer Learning** – Introduces sequence-to-sequence models, attention mechanisms, self-attention, transformers (BERT, GPT-2), and their applications in advanced NLP tasks like text summarization.

Expected Learning Outcomes

- Understand key concepts and techniques in Natural Language Processing (NLP).
- Be proficient in using NLP libraries such as NLTK, SpaCy, and TextBlob for text processing and analysis.
- Learn to build and train sequential models (RNN, LSTM, GRU) for text-based tasks.
- Implement transformer-based models like BERT and GPT-2 for advanced NLP applications.
- Gain hands-on experience in solving real-world NLP problems, including news categorization, emotion detection, and text summarization.

Milestone 6: Generative AI | Week: 33-36

Milestone description

This milestone introduces learners to Generative AI, focusing on text generation, machine translation, large language models (LLMs), Langchain, vector databases, Hugging Face, prompt engineering, retrieval-augmented generation (RAG), and fine-tuning. Through hands-on projects, learners will build AI-powered assistants, content retrieval systems, and creative writing models while gaining expertise in modern AI frameworks and ethical considerations.

Comprehensive List of Topics

- **Generative AI Intro & Text Generation** – Covers the fundamentals of generative AI, including probabilistic modeling, autoencoders, variational autoencoders (VAEs), and GPT models for text generation. Introduces key challenges like controlling AI-generated content and ensuring coherent outputs, with hands-on implementation in autoencoder-based poetry generation.
- **Generative AI for Machine Translation** – Explores traditional and deep learning-based translation methods (SMT, NMT), attention mechanisms, and GPTs for machine translation. Covers AI applications in multilingual text generation, poetry, music composition, and ethical considerations in creative AI, with a project on ethical bias detection in machine translation.
- **LLM & Langchain Framework** – Introduces Large Language Models (LLMs), their industry applications, and integrations with LangChain and LlamaIndex for structured data-driven AI assistants.
- **Vector Databases (Vector DB)** – Covers vector databases, their importance, types (e.g., ChromaDB), and implementation for AI-powered search and retrieval systems.
- **Hugging Face & Ollama** – Explores Hugging Face's pre-trained models, real-world applications, and Ollama's role in creating and fine-tuning custom LLM models.
- **Prompt Engineering & Retrieval-Augmented Generation (RAG)** – Focuses on crafting effective prompts, advanced RAG techniques, and frameworks like LangChain for AI-powered text summarization, QA, and creative writing.
- **Fine-Tuning LLMs** – Covers different fine-tuning approaches for LLMs, including real-world applications, domain-specific adaptations, and optimizing AI-generated outputs.

Expected Learning Outcomes

- Gain expertise in generative AI techniques for text generation, translation, and storytelling.
- Learn advanced NLP models, including GPT, VAEs, transformers, and multimodal models.
- Work with LLMs and LangChain for context-aware AI applications.
- Implement vector databases (VectorDB) for intelligent search and retrieval.
- Utilize Hugging Face & Ollama for deploying generative AI models.
- Master prompt engineering and RAG for efficient AI responses.

- Learn fine-tuning techniques for customizing LLMs for specific applications.

Projects

Project	Description
Sign Language Detection	This project builds an AI-powered system to detect and interpret sign language from images or videos. It enhances communication accessibility for the hearing impaired.
Named Entity Recognition (NER)	This project develops a Named Entity Recognition (NER) system to identify and classify entities like names, dates, and locations in text. It enhances information extraction and text analysis for various applications.
AI-Powered Custom Chatbot	This project develops a tailored chatbot using GPT, designed to provide domain-specific responses and personalized interactions. It enhances user experience across various applications.
GeneAI – Intelligent Voice Assistant	This project develops GeneAI, a smart voice assistant similar to Alexa, enabling seamless voice interactions and task automation. It enhances user convenience through AI-driven responses and commands.
AI-Powered Image and Caption Generator	This project develops a system that generates images and captions based on user prompts using advanced AI models. It enhances creativity and automation in content generation.